Keep in mind that these are simplified descriptions of sophisticated DSP algorithms; homomorphic processing is filled with subtle details. For example, the logarithm must be able to handle both negative and positive values in the input signal, since this is a characteristic of audio signals. This requires the use of the *complex logarithm*, a more advanced concept than the logarithm used in everyday science and engineering. When the linear filtering is restricted to be a *zero phase* filter, the complex log is found by taking the simple logarithm of the absolute value of the signal. After passing through the zero phase filter, the sign of the original signal is reapplied to the filtered signal.

Another problem is *aliasing* that occurs when the logarithm is taken. For example, imagine digitizing a continuous *sine wave*. In accordance with the sampling theorem, two or more samples per cycle is sufficient. Now consider digitizing the logarithm of this continuous sine wave. The sharp corners require many more samples per cycle to capture the waveform, i.e., to prevent aliasing. The required sampling rate can easily be 100 times as great after the log, as before. Further, it doesn't matter if the logarithm is applied to the continuous signal, or to its digital representation; the result is the same. Aliasing will result unless the sampling rate is high enough to capture the sharp corners produced by the nonlinearity. The result is that audio signals may need to be sampled at 100 kHz or more, instead of only the standard 8 kHz.

Even if these details are handled, there is no guarantee that the linearized signals *can* be separated by the linear filter. This is because the spectra of the linearized signals can overlap, even if the spectra of the original signals do not. For instance, imagine adding two sine waves, one at 1 kHz, and one at 2 kHz. Since these signals do not overlap in the frequency domain, they can be completely separated by linear filtering. Now imagine that these two sine waves are multiplied. Using homomorphic processing, the log is taken of the combined signal, resulting in the log of one sine wave plus the log of the other sine wave. The problem is, the logarithm of a sine wave contains many harmonics. Since the harmonics from the two signals overlap, their complete separation is not possible.

In spite of these obstacles, homomorphic processing teaches an important lesson: signals should be processed in a manner *consistent* with how they are formed. Put another way, the first step in any DSP task is to understand how information is represented in the signals being processed.